Deepankar Karn

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Summary

Results-driven MCA graduate and .NET C# developer with hands-on internship experience at EPAM Systems and a portfolio of four independently built, metrics-backed backend systems. Proficient in system design, data structures & algorithms, microservices, and CI/CD pipelines. Seeking a fresher Software Engineer role where I can contribute to production-grade backend development from day one.

Education

Asansol Engineering College — Asansol, West Bengal
Master of Computer Applications (MCA) – CGPA: 8.09/10 — *Oct 2023 – Jul 2025 (Completed)*

Asansol Engineering College — Asansol, West Bengal
*Bachelor of Computer Applications (BCA) – **CGPA: 9.28/10 (Distinction)*** — *Sep 2020 – Jul 2023*

Experience

Junior Application Support Analyst Intern — Jan 2025 – Jun 2025
EPAM Systems — *Hyderabad, Telangana, India*

- Reduced average critical-incident resolution time by ~30% by independently designing a structured knowledge base of technical runbooks for .NET C# enterprise applications, enabling faster triage by cross-functional Agile Scrum teams.
- Improved team onboarding efficiency by 40% by authoring end-to-end technical documentation covering microservices architecture patterns and CI/CD pipeline workflows (GitHub Actions), cutting analyst ramp-up time from two weeks to under one.
- Prevented recurring production issues across 3 client-facing systems by applying SOLID principles and design pattern analysis (Factory, Singleton) to root-cause diagnosis, delivering preventive fixes validated through unit testing before deployment.
- Reduced deployment errors by 20% by collaborating with senior engineers to review and harden CI/CD pipelines, applying system design principles to identify and eliminate failure points in the build-to-release workflow.

Projects

Finance Data Processing API | *C#, .NET 8, PostgreSQL, EF Core, Docker, JWT, RBAC* (Independent Project)
Built a production-ready secure REST API with JWT/RBAC auth, dynamic LINQ querying, and fully containerised Docker deployment — reducing query latency by ~35% and environment setup time by 95%.

- Reduced unauthorized data access risk to zero by architecting JWT authentication and Role-Based Access Control (RBAC), enforcing strict separation between admin and standard user privileges across all financial ledger endpoints.
- Improved large-dataset query performance by ~35% by engineering dynamic filtering, search, and pagination using LINQ, applying data structures optimized for efficient traversal of high-volume financial records.
- Eliminated data integrity failures by designing a PostgreSQL schema via Entity Framework Core with decimal precision constraints and soft-delete guardrails, modelled after system design principles for audit-proof financial applications.
- Cut environment setup time from hours to under 5 minutes by containerizing the full stack using Docker Compose, enabling reproducible microservices-style deployment across development and production environments.

Autonomous Claims Processing Agent | *C#, .NET 8, Google Gemini AI, REST APIs, xUnit* (Independent Project)
Engineered an AI document pipeline using Google Gemini that cut manual data entry by ~70% across insurance claim processing workflows, with 3× reliability improvement through xUnit test coverage.

- Reduced manual data entry effort by an estimated 70% by engineering an AI-driven pipeline using Google Gemini AI to autonomously extract structured fields from unstructured PDF insurance claims.
- Achieved zero misrouted claims in test scenarios by developing a custom rule-based routing engine applying decision-tree logic to evaluate extracted data and automate multi-stage approval workflows.
- Improved pipeline reliability by 3× by writing unit tests using xUnit, validating edge cases across malformed PDFs and incomplete claim fields before deployment.

Real-Time Inflation-Adjusted SIP Calculator | *C#, .NET, RESTful API, Async/Await, JSON* (Independent Project)

- Delivered accurate 30-year SIP projections with live inflation adjustment by integrating the World Bank CPI API via HttpClient, improving output accuracy over static models by ~25%.
- Eliminated UI blocking entirely by implementing async/await patterns, reducing perceived response latency from ~3 seconds to near-instant during live API calls.
- Reduced invalid-input crashes to zero by engineering a custom exception-handling framework (ZeroAndNegativeException, InvalidYearException) built on algorithmic guard-clause patterns.

Console-Based Banking System | *C#, .NET, Factory Pattern, OOP, SOLID Principles, xUnit* (Independent Project)

- Reduced account creation code complexity by 60% by implementing the Factory Design Pattern, abstracting runtime instantiation of Savings and Current account objects and making the system extensible without modifying core logic.
- Prevented 100% of invalid transactions in test scenarios by engineering a custom exception-handling framework (InsufficientBalanceException, DuplicateAccountNumberException) with business-rule constraints enforced at the data structure level.

Technical Skills

Primary Skills: C#, .NET 8, ASP.NET Core, **Data Structures & Algorithms**, **System Design**
Backend & APIs: RESTful APIs, Entity Framework Core, Microservices, JWT/RBAC, Asynchronous Programming
Testing & Quality: xUnit, Unit Testing, Integration Testing, TDD, SOLID Principles, Design Patterns (Factory, Singleton)
DevOps & Cloud: Docker, Kubernetes, GitHub Actions (CI/CD), AWS (EC2, S3), Git, Netlify
Databases: PostgreSQL, MySQL, MongoDB, Firebase
Languages: C#, JavaScript, SQL, Shell Scripting (Bash)
Tools: Visual Studio, JIRA, Linux/Unix

Leadership

Training & Placement Coordinator During MCA
Asansol Engineering College

- Drove campus recruitment outcomes for the batch by coordinating with multiple companies, managing end-to-end logistics for placement drives, and acting as the primary liaison between recruiters and 100+ student candidates.

Department Representative – Computer Applications 5 Years (BCA + MCA)
Asansol Engineering College

- Represented the department to the NAAC Peer team during institutional accreditation, communicating academic and infrastructure highlights to national evaluators — contributing to the college's accreditation outcome.